Worksheet 1: Fractions

Part A: Numerical Fractions Review (10 problems)

Simplify or evaluate each expression. Show your work.

1. $\frac{3}{4} + \frac{5}{6}$

2. $\frac{7}{8} - \frac{2}{3}$

3. $\frac{5}{9} \times \frac{3}{10}$

4. $\frac{4}{5} \div \frac{2}{15}$

5. $\frac{2}{3} + \frac{1}{4} - \frac{1}{6}$

6. $\frac{7}{12} \times \frac{8}{21} \times \frac{3}{2}$

7. $\frac{\frac{3}{4}}{\frac{9}{16}}$ (complex fraction)

8. $\frac{\frac{2}{5} + \frac{1}{3}}{\frac{1}{2}}$

9. $2\frac{1}{3} \times 1\frac{1}{2}$ (mixed numbers)

10. $4\frac{2}{5} \div 1\frac{1}{10}$

Part B: Algebraic Fractions - Simplification (8 problems)

Simplify each expression completely. Assume no denominators equal zero.

11. $\frac{6x^2}{9x}$

12. $\frac{15a^3b^2}{25ab^4}$

13. $\frac{x^2 - 4}{x + 2}$

14. $\frac{x^2 - 9}{x^2 + 6x + 9}$

15. $\dfrac{2x^2+4x}{x^2+2x}$

16. $\dfrac{x^2-5x+6}{x^2-4}$

17. $\dfrac{6x^2-x-2}{2x^2+3x-2}$

18. $\dfrac{x^3-8}{x^2-4}$

Part C: Algebraic Fractions - Operations (8 problems)

Perform the indicated operations and simplify completely.

19. $\dfrac{2}{x}+\dfrac{3}{x}$

20. $\dfrac{5}{x+1}-\dfrac{2}{x+1}$

21. $\dfrac{3}{x}+\dfrac{2}{y}$

22. $\dfrac{x}{x+2}+\dfrac{2}{x-2}$

23. $\dfrac{1}{x-1}-\dfrac{1}{x+1}$

24. $\dfrac{x}{x^2-1}+\dfrac{1}{x+1}$

25. $\dfrac{2x}{x+3}\times\dfrac{x^2-9}{4x^2}$

26. $\dfrac{x^2-4}{x+5}\div\dfrac{x-2}{x^2+5x}$

Part D: Complex Algebraic Fractions (4 problems)

Simplify each complex fraction.

27. $\dfrac{\frac{1}{x}+\frac{1}{y}}{\frac{1}{x}-\frac{1}{y}}$

28. $\dfrac{\frac{2}{x+1}-\frac{1}{x}}{\frac{3}{x}}$

29. $\dfrac{1-\frac{1}{x^2}}{1+\frac{1}{x}}$

30. $\dfrac{\frac{a}{b}+\frac{b}{a}}{\frac{a^2-b^2}{ab}}$

Part E: Reasoning About Fraction Behavior (6 problems)

Answer each question with a brief explanation. No calculation required.

31. Consider the fraction $\frac{n}{10}$ where n is a positive number. - (a) What happens to the value of the fraction as n gets larger and larger? - (b) What value does the fraction approach as n approaches 0?

32. Consider the fraction $\frac{10}{n}$ where n is a positive number. - (a) What happens to the value as n gets larger and larger? - (b) What happens to the value as n gets closer and closer to 0 (but stays positive)?

33. For the fraction $\frac{x+1}{x}$ where $x > 0$: - (a) Rewrite this as a sum of two simpler fractions. - (b) As x gets very large (like 100, 1000, 10000…), what value does this fraction approach? - (c) Why does this happen?

34. Consider $\frac{x^2}{x^2+1}$ for positive values of x. - (a) Calculate the value when $x = 1$, $x = 10$, and $x = 100$. - (b) What value does this fraction seem to approach as x grows without bound? - (c) Will the fraction ever actually reach that value? Why or why not?

35. You have the fraction $\frac{a}{b}$ where both a and b are positive. - (a) If you double both the numerator and denominator, what happens to the value? - (b) If you add 1 to both the numerator and denominator, does the value increase, decrease, or stay the same? (Hint: test with $\frac{1}{2}$ and $\frac{3}{4}$) - (c) Can you explain the pattern from part (b)?

36. The expression $\frac{1}{1+\frac{1}{x}}$ can be simplified. - (a) Simplify this expression to a single fraction. - (b) As x approaches infinity, what does the simplified expression approach? - (c) As x approaches 0 (from the positive side), what happens to the value?

Part F: Multi-Step Problems (4 problems)

These problems require multiple operations. Show all steps.

37. Simplify completely: $\frac{x^2-1}{x^2+2x} \cdot \frac{x^2+4x+4}{x^2+x-2} \div \frac{x+2}{x}$

38. Express as a single fraction in lowest terms:

$$\frac{1}{x} + \frac{1}{x+1} + \frac{1}{x+2}$$

39. If $\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$, solve for c in terms of a and b.

40. The sum of a number and its reciprocal is $\frac{5}{2}$. - (a) Write an equation representing this situation. - (b) Multiply through to clear fractions. - (c) Find the number(s).